

Attention-Augmented NGCF: Personalized Graph Recommendation via Learnable Neighbor Weighting

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ABSTRACT

Neural Graph Collaborative Filtering (NGCF) [12] has established a strong baseline for graph-based recommendation by modeling high-order user-item interactions. We propose an attention-enhanced NGCF variant that introduces node-specific, learnable neighbor weights via an MLP-based attention mechanism. Our approach improves personalization while retaining multi-hop message-passing efficiency. While our implementation focuses on NGCF, the method is generic and can be adapted to simplified frameworks such as LightGCN [5]. Experiments on the MovieLens-100K dataset [9] show that our attention-enhanced model consistently matches or exceeds the performance of vanilla NGCF, especially in sparse user-item interactions.

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1 INTRODUCTION

Personalized recommendation systems are among the most widely adopted applications of machine learning, with significant impact across e-commerce, advertising, social media, and more. Unlike popularity-based or sponsored recommendations, they leverage users' historical interactions to provide tailored suggestions, enhancing relevance and reducing search effort. This field began with traditional matrix factorization-based methods, but has advanced rapidly with the introduction of modern machine learning paradigms.

Traditional recommendation algorithms often relied on matrix-based user-item representations. However, modern systems extend these foundations by incorporating rich structural signals via knowledge graphs [8], and leveraging advanced learning techniques such as recurrent neural networks, graph neural networks (GNNs), and reinforcement learning [10]. This evolution reflects an industry-wide push toward more accurate, dynamic, and scalable recommender systems suitable for real-world deployments. In

parallel, recent efforts have also explored the integration of **generative models**, particularly large language models (LLMs), into the recommendation pipeline [1]. These models not only improve recommendation quality but also transform user interaction with systems—enabling personalized narratives, conversational interfaces, and adaptive content. This growing interest in generative approaches signals a paradigm shift in how personalization is achieved in state-of-the-art recommendation systems.

Neural graph-based models, known for their simplicity and effectiveness, have emerged as powerful alternatives by integrating both content and structural information, enabling them to model higher-order relations and provide more accurate, context-aware recommendations. Building upon these graph-based foundations, recent advancements in recommendation systems increasingly incorporate additional contextual signals—such as temporal dynamics, knowledge graphs, and multi-modal information—to further enhance personalization. Several state-of-the-art models, including TGCL4SR for temporal contrastive learning [3], M3KGR for multi-modal KG reasoning [6], and KAUR for contrastive user embeddings [4], exemplify this trend. Meanwhile, novel directions like Spectro-Riemannian GNNs [7] highlight the potential of learning representations over curved graph manifolds, opening new avenues for capturing complex relational structures in recommendations.

Nevertheless, graph-based collaborative filtering models leveraging neural message passing, especially NGCF [12] and LightGCN [5], remain central to the field due to their scalability and effectiveness in capturing higher-order connectivity patterns. These models continue to serve as strong baselines and key components in many of the latest recommendation approaches.

NGCF (Neural Graph Collaborative Filtering) pioneered the use of GCN-based propagation in recommendation by modeling explicit high-order relations. LightGCN [5] then proposed simplifying the architecture by removing feature transformations and nonlinearities, resulting in a lightweight yet strong baseline. Despite its simplicity, LightGCN has become the de facto standard in neural graph recommendation research thanks to its impressive performance and efficiency.

However, recent works have started to surpass LightGCN by exploring more expressive techniques, including graph contrastive learning methods such as LightGCL [15] and SimGCL [13], or neural ODE-based recommenders like CDE-CF [14]. These newer models often achieve higher accuracy (e.g., in NDCG or Recall) and greater robustness on benchmark datasets. Nevertheless, most of these advancements continue to use NGCF-like techniques or LightGCN as their benchmark baselines, underlining the continued significance of these architectures. We hypothesize that an attention-augmented

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method could further improve these classical models and make them competitive with recent state-of-the-art methods.

While NGCF and LightGCN remain popular for their ability to model high-order relations, both rely on fixed normalization in neighbor aggregation, which limits their ability to adapt to the diverse importance of neighbors for different nodes. Prior work such as GACSE [2] introduced attention mechanisms in recommender systems by designing complex heterogeneous neighbor modeling pipelines. In contrast, our approach directly augments NGCF's aggregation function with a simple, learnable attention mechanism, enabling dynamic, node-specific weighting of neighbors during message propagation. Our method is generic, lightweight, and can also be adapted to simplified frameworks like LightGCN. We demonstrate experimental results using NGCF as the baseline to show the benefits of our attention-enhanced formulation.

2 MODEL FORMULATION: NEURAL GRAPH COLLABORATIVE FILTERING WITH ATTENTION

In this section, we present the foundational NGCF [12] architecture and describe our attention-enhanced extension that improves neighbor aggregation by learning personalized importance scores. We formalize the graph structure, message construction, and embedding propagation used in our framework.

Author Note

All novel contributions introduced in this work are **box highlighted in blue** or **blue textcolor**, including attention-based message passing and model adaptations. These highlight our extensions over the original NGCF framework.

2.1 Neural Graph Collaborative Filtering (NGCF) Key Formulation

We consider a bipartite graph $G_r = (U, I, E)$, where $U = \{u_1, \dots, u_N\}$, $I = \{i_1, \dots, i_M\}$, and edges $E \subseteq U \times I$ represent user-item interactions. Each node has an embedding vector $\mathbf{e}_u, \mathbf{e}_i \in \mathbb{R}^d$, initialized randomly or using side attributes.

Message Construction and Propagation

At layer l , the embedding of user u is:

$$\mathbf{e}_u^{(l)} = \sigma \left(\mathbf{m}_{u \leftarrow u}^{(l)} + \sum_{i \in \mathcal{N}_u} \mathbf{m}_{u \leftarrow i}^{(l)} \right), \quad (1)$$

where σ is the LeakyReLU activation. The message from item i to user u is:

$$\mathbf{m}_{u \leftarrow i}^{(l)} = \frac{1}{\sqrt{|\mathcal{N}_u| |\mathcal{N}_i|}} \left(\mathbf{W}_1^{(l)} \mathbf{e}_i^{(l-1)} + \mathbf{W}_2^{(l)} (\mathbf{e}_i^{(l-1)} \odot \mathbf{e}_u^{(l-1)}) \right), \quad (2)$$

and the self-message:

$$\mathbf{m}_{u \leftarrow u}^{(l)} = \mathbf{W}_1^{(l)} \mathbf{e}_u^{(l-1)}. \quad (3)$$

Matrix Propagation

NGCF propagation in matrix form:

$$\mathbf{E}^{(l)} = \sigma \left((\mathbf{L} + \mathbf{I}) \mathbf{E}^{(l-1)} \mathbf{W}_1^{(l)} + (\mathbf{L} \mathbf{E}^{(l-1)} \odot \mathbf{E}^{(l-1)}) \mathbf{W}_2^{(l)} \right), \quad (4)$$

where $\mathbf{L} = \tilde{\mathbf{D}}^{-1/2} \tilde{\mathbf{A}} \tilde{\mathbf{D}}^{-1/2}$ is the normalized adjacency matrix.

2.2 Attention-Augmented Message Construction (Our Work)

Unified Interpretation. Our attention-augmented NGCF combines:

- **Laplacian-based propagation:** Encodes high-order graph structure through spectral smoothing.
- **Attention-based aggregation:** Captures semantic neighbor importance within each user/item's local 1-hop neighborhood.

While the Laplacian operator provides global smoothing and propagation, the attention mechanism allows fine-grained control over local influence, especially when attributes are included or neighborhood quality varies.

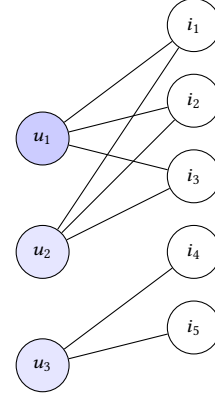


Figure 1: NGCF: High-order connectivity via Laplacian propagation.

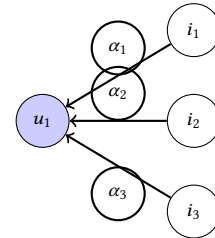


Figure 2: Our Attention-enhanced NGCF: Neighbor contributions weighted by $\alpha_{u,i}$.

We enhance NGCF with an attention mechanism [11] that learns the importance of each neighbor during aggregation. Instead of using fixed normalization, we learn attention weights $\alpha_{u,i}$ to adaptively weight neighbors.

User Embedding Update Rule

$$\mathbf{e}_u^{(l)} = \sigma \left(\mathbf{m}_{u \leftarrow u}^{(l)} + \sum_{i \in \mathcal{N}_u} \mathbf{m}_{u \leftarrow i}^{(l)} \right) \quad (5)$$

Attention-Based Message Definitions (Layer l)

$$\mathbf{m}_{u \leftarrow i}^{(l)} = \rho_{ui} \left(\mathbf{W}_1^{(l)} \mathbf{e}_i^{*(l-1)} + \mathbf{W}_2^{(l)} \left(\mathbf{e}_i^{*(l-1)} \odot \mathbf{e}_u^{*(l-1)} \right) \right) \quad (6)$$

$$\mathbf{m}_{u \leftarrow u}^{(l)} = \mathbf{W}_1^{(l)} \mathbf{e}_u^{*(l-1)} \quad (7)$$

User-Side Aggregation with Attention (Layer $l - 1$ Embeddings)

$$\mathbf{e}_u^{*(l-1)} = \sum_{i \in \mathcal{N}_u} \alpha_{u,i}^{(l)} \cdot \mathbf{e}_i^{*(l-1)} \quad (8)$$

where the attention weight $\alpha_{u,i}^{(l)}$ is computed as:

$$\alpha_{u,i}^{(l)} = \frac{\exp \left(\mathbf{v}_a^\top \tanh \left(\mathbf{W}_a [\mathbf{e}_u^{*(l-1)} \parallel \mathbf{e}_i^{*(l-1)}] \right) \right)}{\sum_{j \in \mathcal{N}_u} \exp \left(\mathbf{v}_a^\top \tanh \left(\mathbf{W}_a [\mathbf{e}_u^{*(l-1)} \parallel \mathbf{e}_j^{*(l-1)}] \right) \right)} \quad (9)$$

Integration into NGCF Propagation Rule

At every layer the attention enriched $\mathbf{E}^{*(l-1)}$ replaces $\mathbf{E}^{(l-1)}$ during the aggregation as compared to NGCF

$$\mathbf{E}^{(l)} = \sigma \left((\mathbf{L} + \mathbf{I}) \mathbf{E}^{*(l-1)} \mathbf{W}_1^{(l)} + \left(\mathbf{L} \mathbf{E}^{*(l-1)} \odot (\mathbf{E}^{*(l-1)}) \right) \mathbf{W}_2^{(l)} \right) \quad (10)$$

Here, $\mathbf{L}$ is the normalized Laplacian matrix, and $\odot$ denotes element-wise multiplication. This formulation allows attention-enhanced embeddings to flow into subsequent layers with richer semantics.

Intuition

- Learns soft relevance scores between center and neighbors.
- Avoids noisy or uninformative neighbors.
- Captures personalized aggregation patterns.

3 IMPLEMENTATION HIGHLIGHTS AND EXPERIMENTAL RESULTS

Algorithm: Attention-Based Aggregation

Our attention mechanism adapts ideas from GAT [11] to the collaborative filtering setting. The algorithm to aggregate embeddings for a node i is summarized below:

Algorithm 1 Attention-based Aggregation for Node i

Require: Embeddings $\{e_j\}$, center node embedding e_i , neighbors $\mathcal{N}(i)$, attention weights $\mathbf{W}_a, \mathbf{v}_a$

- 1: **for** each $j \in \mathcal{N}(i)$ **do**
- 2: Concatenate features: $\mathbf{c}_{ij} = [e_i \parallel e_j]$
- 3: Compute hidden representation: $\mathbf{s}_{ij} = \tanh(\mathbf{W}_a \mathbf{c}_{ij})$
- 4: Compute raw score: $\alpha_{ij} = \mathbf{v}_a^\top \mathbf{s}_{ij}$
- 5: **end for**
- 6: Normalize scores: $\alpha_{ij} = \text{softmax}_j(\alpha_{ij})$
- 7: Aggregate embedding: $e_i^{\text{agg}} = \sum_{j \in \mathcal{N}(i)} \alpha_{ij} e_j$
- 8: **return** e_i^{agg}

Efficient PyTorch Implementation

Our attention-based aggregation is implemented efficiently in PyTorch using batched matrix operations:

- Neighbor and center embeddings are collected via preprocessed index matrices.
- Pairwise attention features are constructed by concatenating center and neighbor embeddings.
- Scores are computed through a two-layer feedforward network and normalized with softmax along the neighborhood dimension.
- Weighted aggregation uses batched matrix multiplication for high computational efficiency on GPUs.

Our implementation, of modified NGCF architecture with attention, is available on [GitHub](#)

Experiments on the MovieLens-100K Dataset

To evaluate our model, we used the MovieLens-100K dataset [9], which provides user-movie interaction records along with user metadata and movie genres. Our approach introduces minor changes to incorporate user and item attributes/features into the learning process—something the original NGCF does not consider. All embeddings were Xavier-initialized as in the original NGCF paper.

Data Preparation. User Features: User metadata includes age, gender, and occupation, encoded as binary one-hot vectors:

```
user_id\tage\tgender\toccupation\tzip\tage_group
1\t24\tM\ttechnician\t85711\t2
```

Example of one-hot encoded features:

```
age_0 age_1 age_2 age_3 age_4 ...
0      0      1      0      0 ...
```

Item (Movie) Features: Movie genres are encoded as one-hot vectors across 19 genres:

```
|Toy Story (1995)|...|0|0|0|1|1|1|...
|GoldenEye (1995)|...|0|1|1|0|0|0|...
```

Evaluation Metric. We used NDCG@10 (Normalized Discounted Cumulative Gain at rank 10) as our primary evaluation metric to compare model configurations and measure ranking performance.

Statistic	Value
# Users	943
# Movies	1682
User features shape	(943, 52)
Item features shape	(1682, 52)

Table 1: Statistics of MovieLens dataset with user and item features

Graph Setup. We built a bipartite graph of users and items with the following properties:

Graph Property	Value
# Users	943
# Items	1682
# Interactions	97,782
Train/Test split	80,808 / 16,974
Sparsity	0.06165
Adjacency matrix size	(2625, 2625)

Table 2: Graph setup and properties for NGCF experiments

Performance Evaluation

We conduct various experiments including user and item attributes and without attributes, evaluated NGCF with and without our attention mechanism across four initialization strategies:

- **Attr. Init. + Frozen:** Attribute-based embeddings, fixed during training.
- **Attr. Init. + Updated:** Attribute-based embeddings, updated during training.
- **Rand. Init. + Frozen:** Random embeddings, fixed during training.
- **Rand. Init. + Updated:** Random embeddings, updated during training.

Table 3: NGCF Performance With/Without Attention (NDCG)

Config	No Attention	With Attention
Attr. Init. + Frozen	0.147	0.143
Attr. Init. + Updated	0.216	0.216
Rand. Init. + Frozen	0.238	0.223
Rand. Init. + Updated	0.231	0.236

Key Observations.

- Attention improves performance when embeddings are updated during training, with NDCG rising to 0.236 for randomly initialized embeddings.
- Attribute-based updated embeddings achieve stable performance, indicating richer semantics when combined with attention.
- Attention may underperform when embeddings are frozen, suggesting limited benefit if representation capacity is constrained.

Limitations and Next Steps

Although our experiments show promising improvements, they are limited to the relatively small MovieLens-100K dataset. Moving forward, we plan to:

- Scale experiments to larger datasets, such as MovieLens-1M, Amazon, and Yelp.
- Perform extensive hyperparameter tuning for better optimization.
- Benchmark against recent state-of-the-art methods, including LightGCL [15] and CDE-CF [14].
- Evaluate the impact of attention on LightGCN; adapting our method to LightGCN should be straightforward.
- Explore more advanced attention mechanisms, such as multi-head attention and dynamic masking.
- Study training stability and robustness in highly sparse recommendation scenarios.

4 CONCLUSION

In this work, we presented an attention-enhanced Neural Graph Collaborative Filtering (NGCF) model that replaces fixed neighbor normalization with learnable, node-specific attention weights. This formulation enables more expressive and personalized message passing while retaining the scalability and multi-hop propagation strengths of NGCF.

Our experiments show that dynamic neighbor weighting with attention can match or outperform standard NGCF configurations across different embedding initialization strategies, demonstrating the potential of adaptive aggregation for improving recommendation quality.

A limitation of our current implementation is the use of fixed-size neighbor sampling (via padding or trimming to constant k), which, while efficient for vectorized computation, introduces challenges:

- Wastes compute on dummy neighbors for sparse nodes,
- Prevents dynamic attention masking with variable neighborhood sizes,
- Could bias learning toward denser users/items.

Mask-aware implementations (e.g., those in PyTorch Geometric or Deep Graph Library) offer potential solutions but are more complex to integrate.

Looking ahead, extending attention mechanisms to multi-hop neighbors, incorporating node-type-specific attention (e.g., distinguishing users from items), and adapting recent advances in GNN architectures will be key directions for future work. Rigorous benchmarking across diverse datasets and tasks—such as cold-start recommendation—will also be important to assess the generalization capabilities of the proposed model.

Given the ongoing evolution of neural graph recommenders, including recent state-of-the-art models, our attention-enhanced NGCF aligns well with emerging trends in personalized recommendation. Our formulation provides a flexible foundation for future research and can inspire attention-augmented variants of related architectures such as LightGCN. We believe continued exploration of learnable neighbor weighting will further advance the performance and adaptability of neural graph-based recommender systems.

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